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In [1]: import pandas as pd # Pandas is a library for data manipulation and analysis
import matplotlib.pyplot as plt # Matplotlib is a plotting library for creat
import seaborn as sns # Seaborn is a library built on top of Matplotlib that

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets

print("--- First 5 Rows ---")
print(df.head()) # prints the first 5 rows of the DataFrame. This is useful

print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many n
# Notice that 'Age' and 'Cabin' have missing values!
df.info() # gives the column names, the number of non-null values in each co
```

--- First 5 Rows ---

	PassengerId	Survived	Pclass	\ 0
1	0	3		
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\ 0
Braund, Mr. Owen Harris	male	22.0	1		
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked	0
0	A/5 21171	7.2500	NaN	S		
1	0	PC 17599	71.2833	C85	C	
2	0	STON/O2. 3101282	7.9250	NaN	S	
3	0	113803	53.1000	C123	S	
4	0	373450	8.0500	NaN	S	

--- Basic Info ---

<class 'pandas.DataFrame'>

RangeIndex: 891 entries, 0 to 890 Data
columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	str
4	Sex	891 non-null	str
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	str
9	Fare	891 non-null	float64
10	Cabin	204 non-null	str

11Embarked 889 non-null str dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

```
In [2]: # Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe()) # prints summary statistics for numerical columns, incl

print("--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")

print(f"The overall survival rate was {df['Survived'].mean():.1%}.") # print
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max}
```

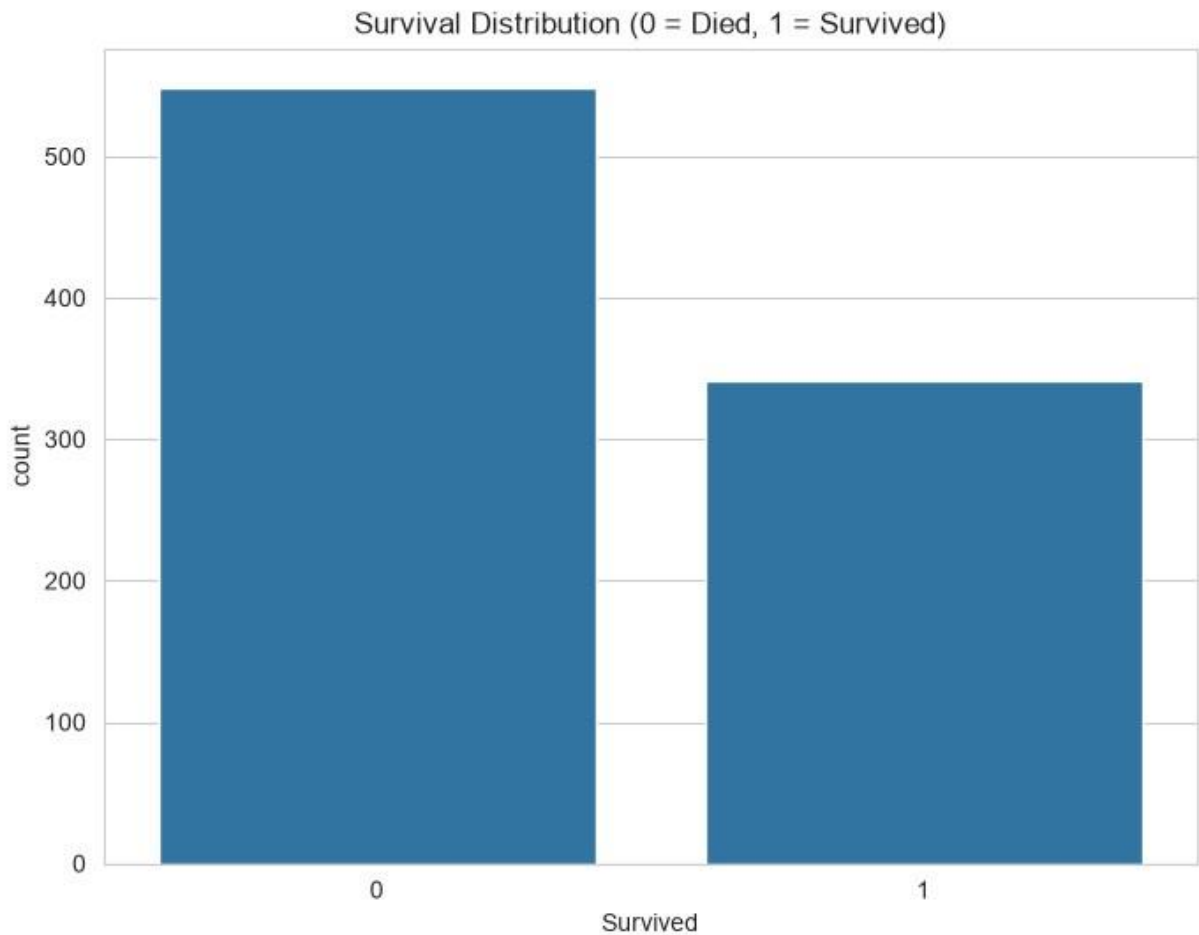
```
--- Descriptive Statistics ---
      PassengerId  Survived  Pclass    Age  SibSp  \
count  891.000000  891.000000  891.000000  714.000000  891.000000  mean
446.000000    0.383838    2.308642   29.699118    0.523008    std
257.353842    0.486592    0.836071   14.526497    1.102743    min
1.000000     0.000000    1.000000    0.420000    0.000000   25%
223.500000    0.000000    2.000000   20.125000    0.000000
50%    446.000000    0.000000    3.000000   28.000000    0.000000   75%
668.500000    1.000000    3.000000   38.000000    1.000000    max
891.000000    1.000000    3.000000   80.000000    8.000000
```

```
      Parch    Fare
count  891.000000  891.000000  mean
0.381594   32.204208  std
0.806057   49.693429  min
0.000000    0.000000  25%
0.000000    7.910400
50%    0.000000   14.454200  75%
0.000000   31.000000  max    6.000000
512.329200

--- Key Insights from Statistics ---
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.0 to a whopping $512.3292.
```

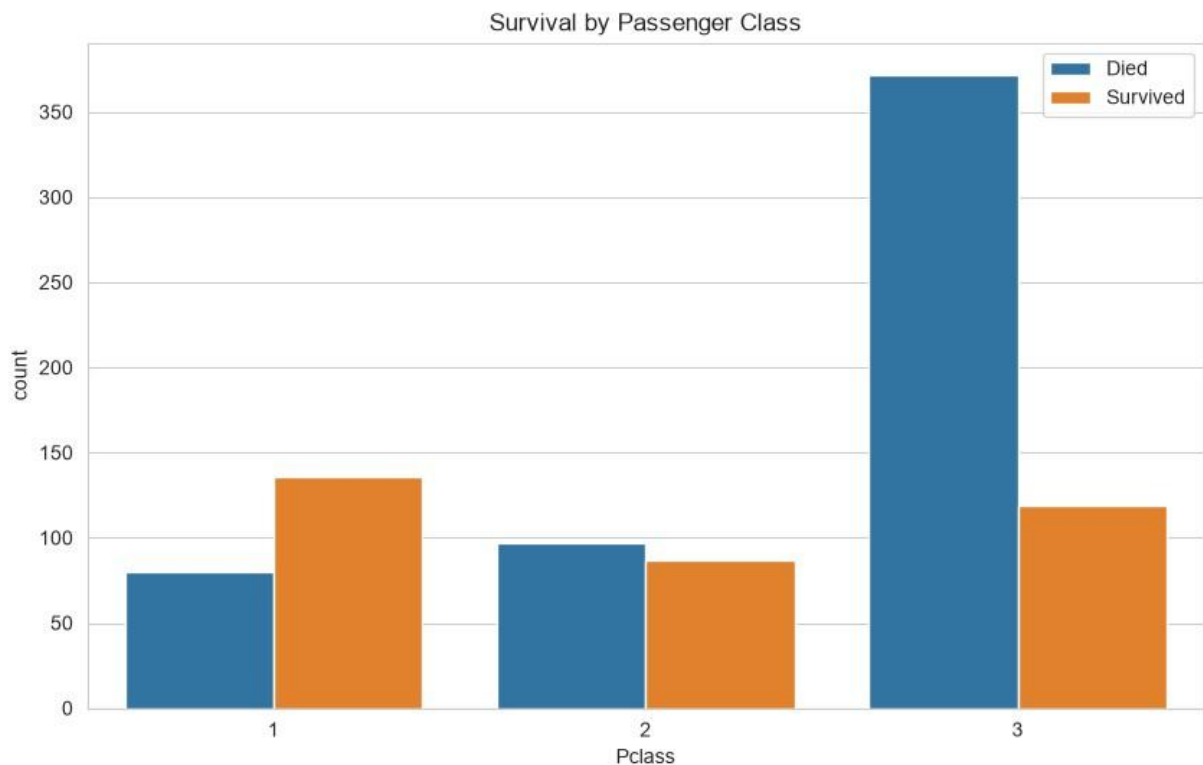
```
In [3]: sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(8, 6)) # Creates a new figure with a specified size (wid
sns.countplot(x='Survived', data=df) # Creates a count plot (bar plot) showi
plt.title('Survival Distribution (0 = Died, 1 = Survived)') # Adds a title t
plt.show() # Displays the plot. This is important in Jupyter notebooks to re
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```
print("Insight: Far more people died than survived. This is an example of an
```



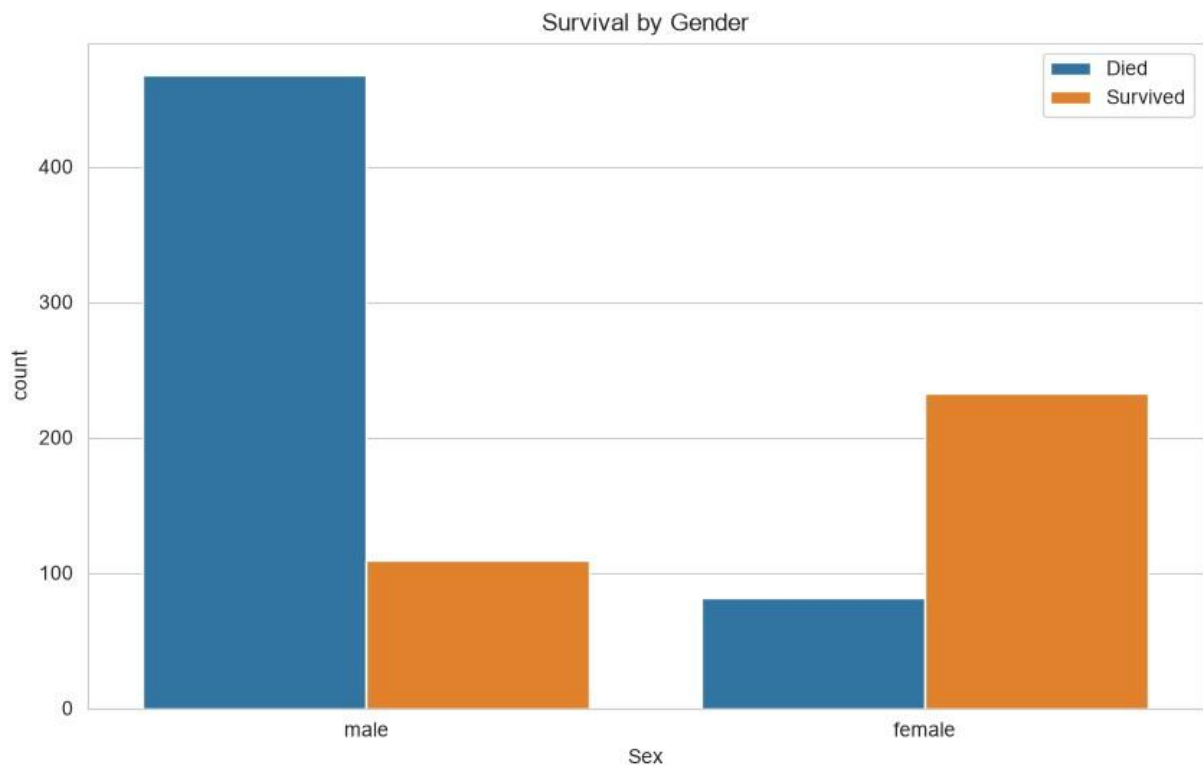
Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for machine learning models.

```
In [ ]: plt.figure(figsize=(10, 6)) sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class') plt.legend(['Died', 'Survived']) #
Adds a legend to the plot, labeling the c plt.show() print("Insight: This is a
very strong pattern. 1st class passengers had a mu
```



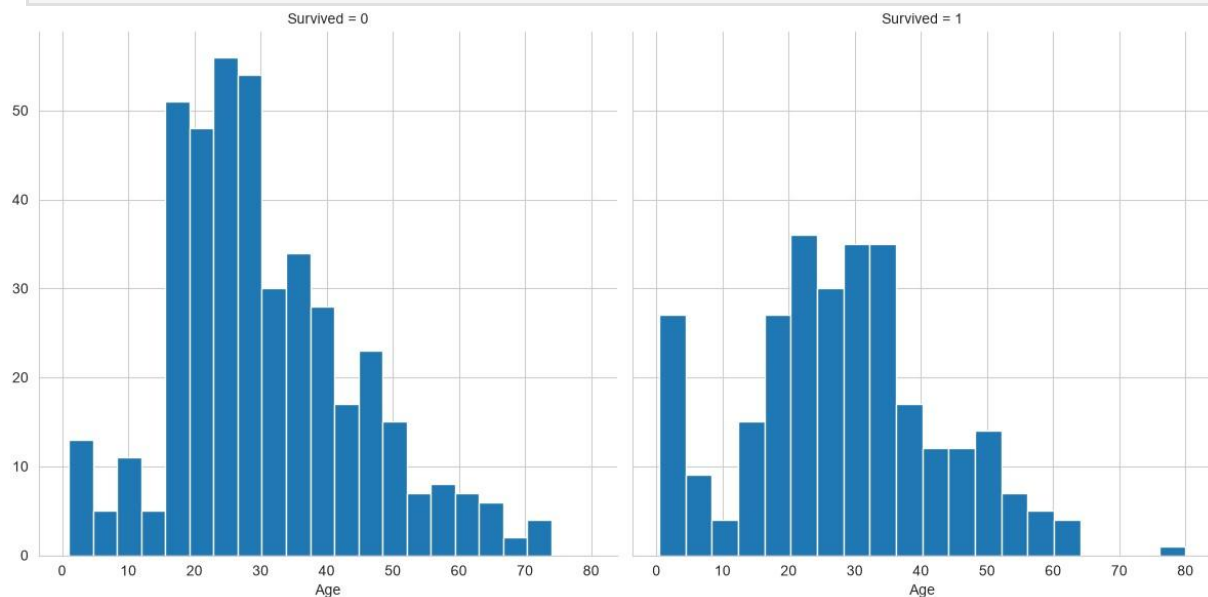
Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers. Money seems to have made a difference.

```
In [ ]: plt.figure(figsize=(10, 6)) sns.countplot(x='Sex', hue='Survived', data=df) #  
Creates a count plot showing plt.title('Survival by Gender') # Adds a title to  
the plot, indicating that plt.legend(['Died', 'Survived']) # Adds a legend to  
the plot, labeling the plt.show() print("Insight: The pattern is undeniable. A  
much higher proportion of female
```



Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong predictor.

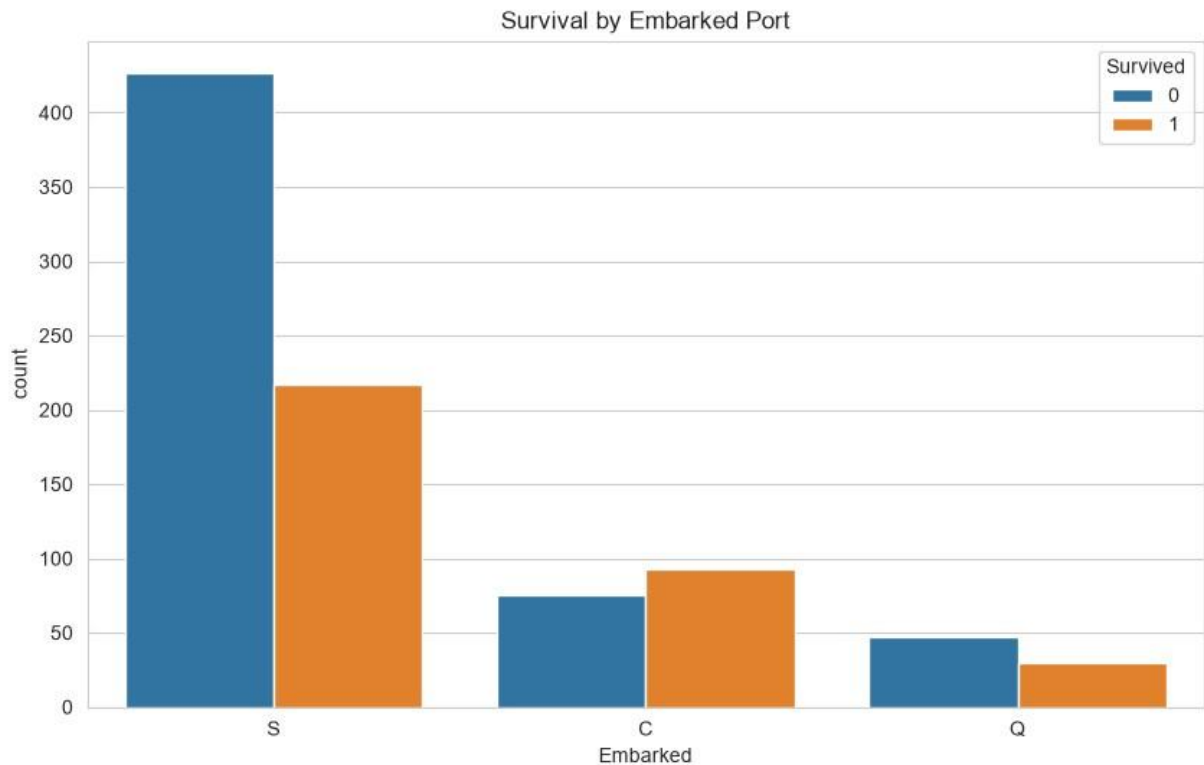
```
In [10]: # A FacetGrid allows us to create multiple plots side-by-side to compare distributions.
# Here, we create one histogram for passengers who died (col='Survived'=0) and one for those who survived (col='Survived'=1).
sns.FacetGrid(df, col='Survived', height=6) # Creates a FacetGrid object
g.map(plt.hist, 'Age', bins=20) # Maps a histogram of the 'Age' column onto each facet
plt.show() # Displays the FacetGrid with the histograms for both groups (survived and died)
print("Insight: The age distribution for those who did not survive is centered around 25-30 years old, while those who survived are more spread out, with a peak around 25-30 years old.")
```



Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, there is a noticeable spike for young children. This supports the 'children' part of the mantra.

1.

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In [ ]: # --- ENTER YOUR CODE HERE ---
sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(10, 6)) # Creates a new figure with a specified size (
sns.countplot(x="Embarked", hue="Survived", data=df) # Creates a count plot
plt.title("Survival by Embarked Port") # Adds a title to the plot, indicatin
plt.show()
```



1.

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In [ ]: # --- ENTER YOUR CODE HERE ---
sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(10, 6)) # Creates a new figure with a specified size
sns.boxplot(x="Survived", y="Fare", data=df) # Creates a box plot showing th
plt.title("Fare Distribution by Survival Status") # Adds a title to the plot
plt.show() # Displays the plot. This is important in Jupyter notebooks to re
```

